



TRADE SHOCKS AND INDUSTRY PERFORMANCES ANALYSIS

GROUP 12 - FOUNDATIONS OF DATA SCIENCE



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Date : December 15, 2025



TRADE SHOCKS AND INDUSTRY PERFORMANCES ANALYSIS REPORT

Objectives

Goals of the Analysis

The primary goal of this study is to evaluate how international trade shocks and exchange rate fluctuations have influenced the financial performance of Canadian firms between 2013 and 2023. By integrating firm-level financial data with industry-level trade exposure, GDP, and exchange rate observations, the analysis seeks to identify which industries and regions are most vulnerable to global trade dynamics, and how firm size shapes resilience.

In addition to these research goals, this project also aims to apply the knowledge gained during the Fundamentals of Science course. By using the basic analytical tools introduced in class, the study provides an opportunity to strengthen data analysis skills and develop a structured workflow. This dual focus ensures that the project not only produces management-ready insights but also supports the development of analytical competencies that will be valuable in future career paths.

Research Questions

To ensure that each team member gains end-to-end hands-on experience, the project is structured around five core research questions:

1. Which industries in Canada are growing, and is there regional variation in performance?
2. Which industries and regions are most exposed to trade shocks?
3. How do international trade shocks impact firm performance across industries?
4. How do CAD-USD exchange rate fluctuations influence profitability in export-heavy versus import-heavy industries?
5. Do small and large firms respond differently to trade shocks within the same industry?

Each question can be assigned to a team member, allowing them to manage the full cycle of analysis - from data preparation and modeling to interpretation and presentation - while still contributing to a unified report

Hypotheses

We hypothesize that industries with greater trade exposure will experience larger swings in profitability during shocks. CAD depreciation is expected to benefit export-heavy industries while compressing margins in import-heavy sectors. Smaller firms are anticipated to be more sensitive to shocks than larger firms, due to limited financial buffers and bargaining power. Regional differences are expected to reflect industry mix, with resource-rich provinces showing greater resilience compared to manufacturing-intensive regions.

Approach

The study will merge firm-level financial datasets (2013–2023) with industry trade exposure, GDP, and exchange rate data. Industry classifications will be harmonized across years, and firms will be segmented by size. The analysis will combine descriptive statistics (growth trends, exposure mapping) with econometric models, including fixed-effects regressions, difference-in-differences, and event studies, to quantify the impact of shocks. Robustness checks will be conducted using alternative exposure metrics and placebo tests to validate findings.

By following this structured approach, the project demonstrates how foundational scientific methods and basic analytical tools can be applied to complex, real-world economic questions -reinforcing skills that will support future professional development and ensuring that each team member gains practical, end-to-end analytical experience.

Data Preparation

Data Sources

Our analysis relied on open government datasets, primarily from the Government of Canada and Statistics Canada. These included:

- **Firm-level financial performance data (2013–2023):** Revenue, expenses, and industry-level ratios, provided in separate CSV files for each year.
- **Trade exposure data:** Industry and province-level measures of import and export exposure.
- **Exchange rate data:** CAD-USD series from Statistics Canada.
- **GDP and trade aggregates:** National trade/GDP ratios to provide macroeconomic context.

All datasets were publicly available and procured directly from official portals. No web scraping or proprietary corporate data was required.

Data Quality and Structure

The datasets were comprehensive, with revenue and expense files containing over 100,000 records. This volume provided strong analytical potential but also introduced challenges in harmonization.

- **Granularity differences:** Revenue files contained province- and industry-level detail, while expense files were only available at the national level.
- **Coverage gaps:** Trade exposure datasets did not include all industries present in the financial data.
- **Yearly fragmentation:** Financial data was split into separate files per year, requiring careful merging and schema alignment.

Despite these challenges, the overall data quality was sufficient for robust analysis, with consistent formatting and reliable metadata.

Tools and Preparation Process

We used Python scripts in Jupiter Notebook to automate downloading, cleaning, and formatting tasks. Key steps included:

- Standardizing column names and data types across years.
- Merging annual financial files into a unified panel dataset.
- Reconciling industry codes (NAICS) to ensure comparability.
- Aligning province-level and national-level datasets where possible.
- Creating visualizations to validate merges and check for anomalies.

Challenges and Team Collaboration

Several blockers arose during preparation:

- **Large file sizes:** Revenue and expense files exceeded 100,000 rows, requiring efficient handling and memory management.
- **Schema inconsistencies:** Each CSV file had slightly different column structures, demanding manual inspection and harmonization.
- **Granularity mismatches:** Province-level detail was available for revenue but not for expenses, limiting certain comparisons.
- **Missing data:** One file was discovered to have incomplete coverage for several years, which required late-night troubleshooting and re-merging.

Our team addressed these challenges collaboratively. Meetings were scheduled regularly, though coordinating during working days was difficult. To overcome this, we adopted a “SWAT team” approach: whenever a blocker was posted in the group chat, whoever was available responded quickly to resolve it. For example, when a file failed to load and merge properly, the issue was flagged late at night and resolved within hours through collective effort.

Outcome

Through active participation from all members, we successfully prepared a unified dataset covering 2013–2023. While some industries and provinces lacked complete data, we proceeded with the available information to ensure timely completion of the report. The preparation process not only enabled the analysis but also strengthened the team’s technical skills in data wrangling, troubleshooting, and collaborative problem-solving.

Analysis

Overview

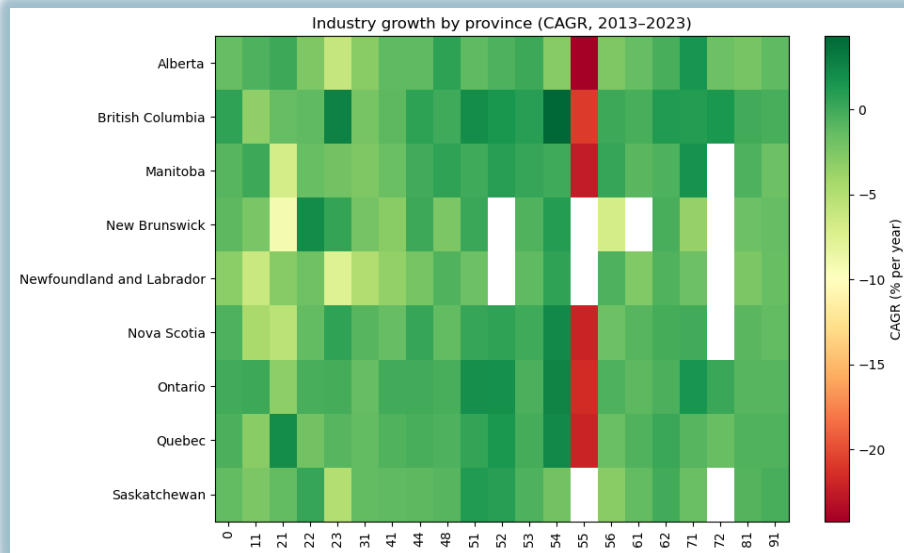
The analysis examines how trade shocks and exchange rate fluctuations affected Canadian industries between 2013 and 2023. Each team member focused on one research question, ensuring end-to-end ownership of data preparation, modeling, and visualization. Together, the findings provide a comprehensive view of industry growth, exposure, and firm-level resilience

1. Industry Growth and Regional Variation

- **Purpose:** Determine which industries are growing in Canada and whether there is regional variation.
- **Analysis:**
 - We use annual data and compute growth metric that reflects real changes, not just inflation:

$$\text{Real GDP} = \text{Nominal GDP} * 100 / \text{CPI}$$

- We use CAGR (Compound Annual Growth Rate) as a growth metric.
- Based of figure 1:



▪ **Biggest decline:** NAICS 55 (Management of companies) dropped sharply across all provinces (~ -20% CAGR).

▪ **Consistent growth:** Finance (52), Professional services (54), Health (62), and Public administration (91) show steady positive growth everywhere.

▪ **Resource provinces:** Alberta, Saskatchewan, and Newfoundland & Labrador grew in mining/oil & gas (21) and transportation (48-49).

▪ **Large provinces:** Ontario, Quebec, and BC have broad service-led growth (retail, info & culture, professional services).

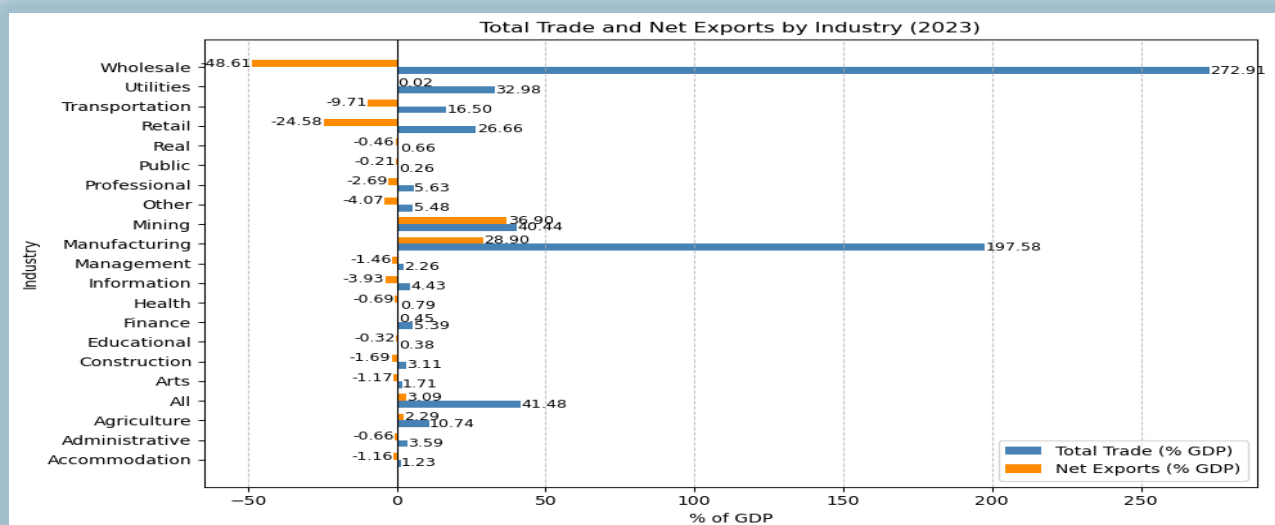
▪ **Data gaps:** White cells indicate missing data - interpret rankings cautiously

- **Key Insight:** Services and public sectors are strong; resource sectors are cyclical; NAICS 55 is a structural decline.

2. Industry and Regional Exposure to Trade Shocks

- **Purpose:** Determine which industries and regions are most exposed to international trade shocks.
- **Analysis:**

Figure 1: Total Trade and Net Exports by Industry



This chart shows (figure 1) the relative size of total trade (% of GDP) and net exports (% of GDP) by industry.

- Blue bars show total trade as percentage of GDP while the orange bars show net exports as percentage of GDP.
- Industries with bars extending far to the right are highly trade exposed.
- Negative values (bars extending left) highlight net importers.
- Manufacturing, wholesale and resource extraction dominate trade exposure, while service industries remain less exposed.
- Manufacturing and resource extraction are net exporters, while wholesale and retail are net importers.

Figure 2: Regional Exposure to Trade Shocks

This visualization highlights regional differences in trade exposure.

- The blue color variation shows the total trade as a percentage of income.
- Ontario and New Brunswick have highest trade exposure as a percentage of GDP. The key trade exposure in New Brunswick is their dependence due to Irving oil exports and port activity, while Ontario's trade is heavily dependent on manufacturing industries like automobiles, machinery and primary metals like steel and aluminum.
- Resource-rich provinces (e.g., Alberta, Saskatchewan) show high export dependence.
- Urban centers (e.g., Ontario, Quebec) reveal stronger import reliance, particularly in consumer goods and technology.

Figure 3: Trade over time

- Trade exposure has expanded markedly since 2020, with both exports and imports registering significant growth.

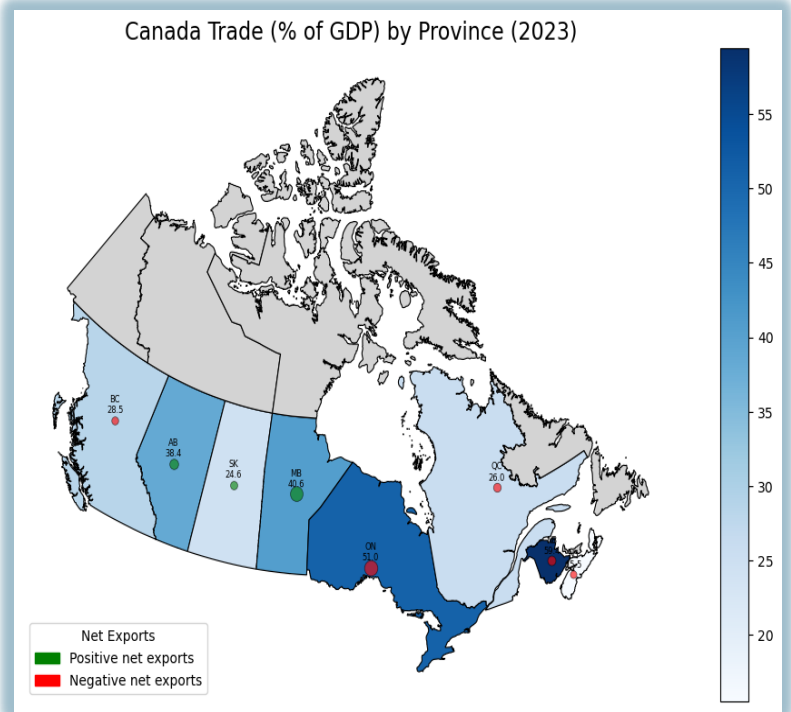


Figure 2: Regional Exposure to Trade Shocks

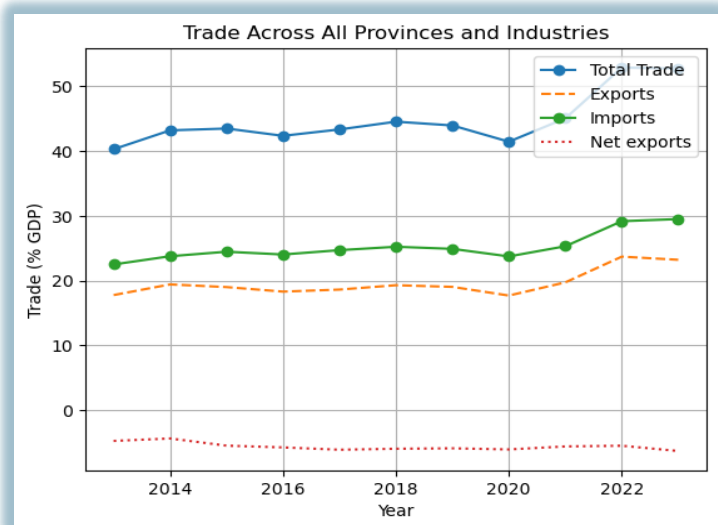


Figure 3: Trade over time

- Net exports have remained persistently negative and exhibit a modest downward trajectory, indicating that the rate of increase in imports has outpaced that of exports.
- **Key Insight:**
 - Trade shocks disproportionately affect industries with high trade-to-GDP ratios.
 - Export-heavy sectors are at risk from global demand downturns, while import-dependent sectors are vulnerable to supply chain disruptions.
 - Regional exposure varies significantly, depending on industrial mix.

3. Impact of Trade Shocks on Firm Performance

- **Purpose:** Measure how trade shocks influenced firm and industry financial outcomes to see how different shocks affected industries across Canada and what the implications of the shock were in regard to revenue, and growth using trade shocks as well as trade exposure based on industry.
- **Analysis:**
 - I conducted 2 separate regression analyses, first one being how trade shocks, or trade exposure affect industry revenue levels:
exposure affect industry revenue levels:
"log_rev ~ trade_shock + net_trade_exposure + C(trade_sector) + C(year)"

	Coefficient	Standard Error	z	P> z	[0.025	0.975]
trade_shock	-0.1398	0.157	-0.893	0.372	-0.447	0.167
net_trade_exposure	-9.19e-05	0.000	-0.529	0.597	-0.000	0.000
C(year)[T.2020]	0.2844	0.074	3.818	0.000	0.138	0.430
C(year)[T.2021]	0.3029	0.116	2.612	0.009	0.076	0.530

- Based on this analysis, both trade shock and net trade exposure were not statistically significant.
- A 1 unit increase in the trade shock results in a -14% change in revenue, however based on the p-value of 0.372 this is not statistically significant
- Now, looking at the years 2020, and 2021, since trade shock is not statistically significant, we can see that both the COVID year, and the rebound were statistically significant macro shocks to revenue levels, but not trade shock.
- The second regression analysis I conducted was how revenue growth was affected by trade shocks or trade exposure:
"rev_growth ~ trade_shock + net_trade_exposure + C(trade_sector) + C(year)"

	Coefficient	Standard Error	z	P> z	[0.025	0.975]
trade_shock	-0.0033	0.008	-0.402	0.688	-0.019	0.013
net_trade_exposure	-3.448e-05	3.01e-05	-1.146	0.252	-9.34e-05	2.45e-05

- Both trade shock and net trade exposure were again not statistically significant.
- Next, I looked at how revenue growth as a whole was affected during large trade shocks which I split into positive and negative shocks. The main negative shock being COVID-19, and the main positive shock being the rebound from COVID-19.
- The point of separating the data into normal, positive shock, and negative shock is to focus on looking at the extreme data that in this case was disrupted by the most major trade shocks
- The most major negative shocks from 2013-2023 include different pricing changes, tariffs & trade policy, and mainly COVID

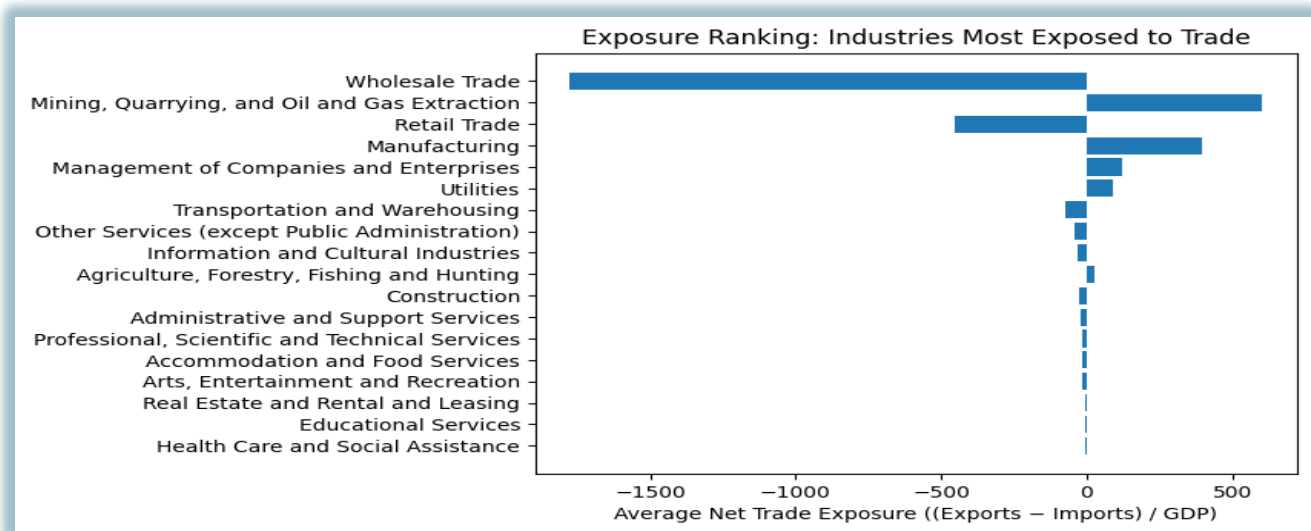
shock_regime	count	shock_regime	mean	median	std	count
Normal	426078	Negative Shock	0.72%	0.000000	270.59%	146396
Positive Shock	147805	Positive Shock	1.48%	0.000000	266.28%	147323
Negative Shock	146595					

- The most major positive shocks include post-COVID rebound, and trade normalization after COVID.
- From the data we can see that the negative trade shocks created a mean revenue growth of 0.72% and positive trade shocks created revenue growth of 1.48%.
- The biggest takeaways from this is that revenue growth is shown to be stable during extreme trade shocks, which is accurate considering our regression analyses results.
- Along with the above revenue growth analysis, I also conducted a sectoral performance ranking during negative trade shock periods, mainly COVID-19 to show the ranking of the different industries' revenue growth or decline when trade was negatively affected.

trade_sector	mean_rev_growth	sector_name
15	62	-0.003604 Health Care and Social Assistance
2	21	-0.003005 Mining, Quarrying, and Oil and Gas Extraction
12	55	-0.002541 Management of Companies and Enterprises
6	41	-0.001384 Wholesale Trade
7	44	-0.000844 Retail Trade
14	61	-0.000253 Educational Services
4	23	-0.000020 Construction
11	54	0.001755 Professional, Scientific and Technical Services
18	81	0.001947 Other Services (except Public Administration)

17	72	0.002480 Accommodation and Food Services
5	31	0.002621 Manufacturing
10	53	0.005228 Real Estate and Rental and Leasing
8	48	0.005654 Transportation and Warehousing
3	22	0.006781 Utilities
9	51	0.012098 Information and Cultural Industries
16	71	0.018344 Arts, Entertainment and Recreation
13	56	0.021011 Administrative and Support Services
1	11	0.093038 Agriculture, Forestry, Fishing and Hunting

- The table above shows us how negative trade shocks affected the different industries.
- The sector that was most negatively affected during trade shocks was Health Care and Social Assistance
- The sector that showed the highest positive revenue growth during negative trade shocks was Agriculture
- Overall, we can see that the trade shock is distributed among the different sectors in different ways which explains why there is no statistical significance of the effect of trade shock on revenue.



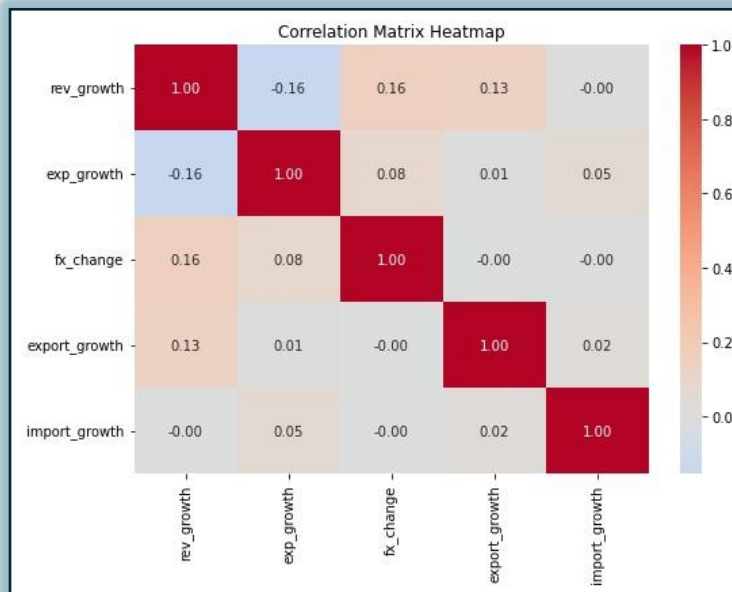
- Last, I looked at how different industries are exposed to trade to get a better outlook on why these trade shocks might have affected different industries in different ways.
- The most important takeaway from this graph is that trade exposure is highly concentrated in a small number of industries with Wholesale trade having the highest negative exposure (largest import) and Mining, Oil, and Gas having the highest positive exposure (largest export).
- **Key Insight:**
 - Across 2013-2023, trade shocks have no statistically significant average effect on industry revenue or revenue growth after controlling for industry structure and macroeconomic conditions.
 - Trade disruptions mainly showed that activity was reallocated across a small number of highly trade-exposed sectors, while economy-wide forces such as COVID-19 dominated overall revenue dynamics.

4. Exchange Rate Fluctuations

- **Purpose:**
Integrates revenue, expenses, exchange rate, and trade data into a unified framework to analyze how Canadian industries respond to CAD–USD exchange rate fluctuations between 2013–2023. It provides insights into profitability, trade shocks, and sensitivity to currency movements.

• Analysis:

Figure 1: Interrelationships Between Growth Metrics and Exchange Rate Changes



- **Revenue vs FX Change (0.16):** Slight positive correlation- revenue tends to grow when CAD weakens.

- **Revenue vs Export Growth (0.13):** Mild positive link -revenue grows modestly with exports

- **Expenses vs FX Change (0.08):** Very weak positive correlation- expenses rise slightly when CAD weakens, but not consistently.

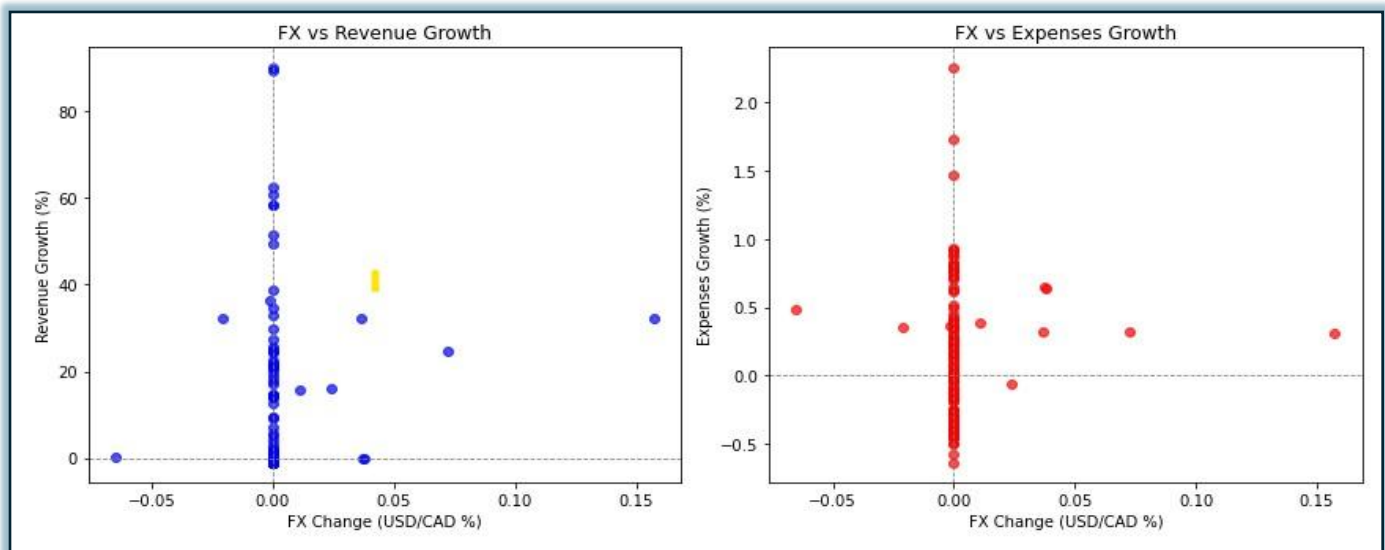
- **Expenses vs Imports (0.05):** Weak link- import growth only slightly affects expenses.

- **FX vs Trade Volumes (~0.05-0.10):** Almost no correlation- exchange rate changes don't strongly drive import/export volumes in this dataset.

- **Revenue vs Expenses (-0.16):** Slight negative correlation - when revenue grows, expenses sometimes shrink, suggesting efficiency gains or cost controls.

In the heatmap, the correlation chart shows that the economic indicators in this dataset move largely independently, with only weak relationships between them. Revenue growth has slight positive links with foreign exchange changes and export growth, and a small negative link with expenditure growth, suggesting modest external influence and possible cost discipline. Expenditure growth itself is mostly unaffected by trade or currency shifts, while exports and imports show almost no connection, pointing to sector-specific dynamics. Overall, the chart highlights that no strong correlations exist, meaning deeper sectoral or time-series analysis is needed to uncover meaningful drivers of performance

Figure 2: Impact of Exchange Rate Changes on Revenue and Expenses Growth



- Left Plot - FX vs Revenue Growth: Shows how changes in the USD/CAD exchange rate relate to revenue growth. Most data points cluster near zero FX change, but there's a slight upward trend-suggesting that when the Canadian dollar weakens (USD/CAD rises), revenue tends to grow. This supports the idea that exporters benefit from currency depreciation.
- Right Plot- FX vs Expenses Growth: Compares FX changes with expense growth. The red dots are tightly clustered, showing very little movement. This means expenses don't respond strongly to exchange rate changes-likely because many costs are domestic or fixed.

When the Canadian dollar gets weaker, company revenues often grow because exporters earn more in Canadian dollars from their U.S. sales. Expenses, however, don't change much with currency movements-they stay stable, which means costs are less affected by exchange rate swings. Overall, the charts suggest that exchange rate changes matter more for revenues than for expenses.

- **Key Insight:**

- Export-heavy industries benefit more from CAD depreciation because revenues rise faster than expenses.
- Import-heavy industries are hurt by CAD depreciation, but the effect is weaker and less consistent.
- FX volatility matters more for revenues than for expenses, making exporters more exposed to currency swings. Shock Years: Outliers in scatter plots reveal periods of unusual behavior (e.g., 2015 CAD depreciation, 2020 pandemic).

5. Firm Size Differences

- **Purpose:**

The purpose of this inquiry is to determine if firm size acts as a differentiating factor in how industries absorb international trade shocks. By isolating the interaction between trade exposure and firm size, we aim to identify if small firms face asymmetric risks compared to their larger counterparts, thereby informing more nuanced industrial policy and support mechanisms

- **Analysis:**

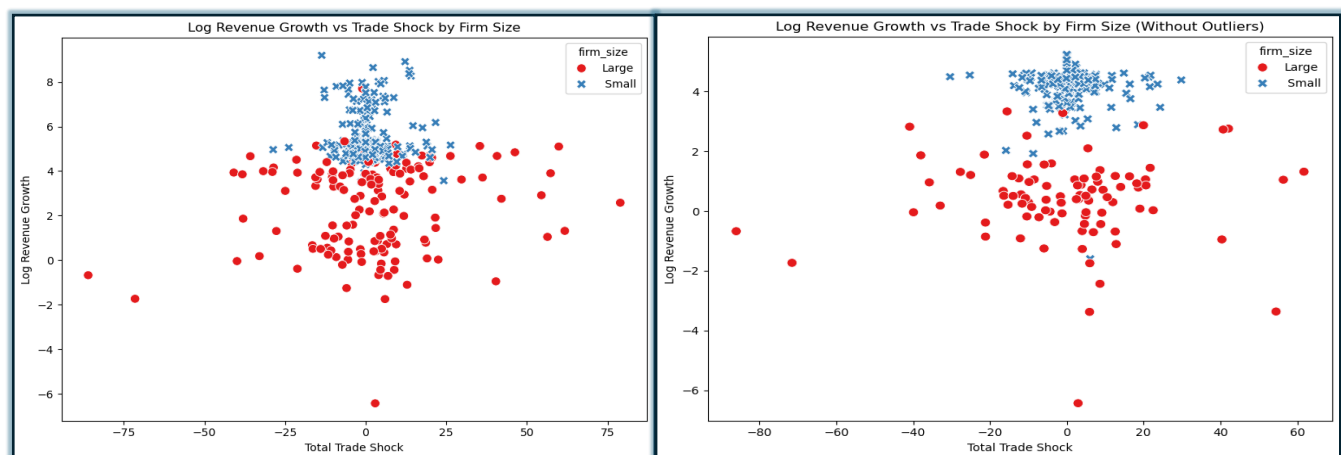
Figure: **Long Revenue Growth vs Trade Shock by firm size** (with and without outliers)

Estimating econometric model

$$y_{it} = \beta_1 * Shock_{it} + \beta_2 * (Shock_{it} * Small_f) + \Gamma X_{it} + \mu_f + \lambda_t + \gamma_i + \varepsilon_{it}$$

y_{it} = Year over year revenue Growth Shock in industry i at time t is calculated using two methods -

- Changes in trade exposure = Percentage change in total trade as % of GDP
 - Change in import penetration = Percentage change in import as % of GDP
 - $Small_f$ = Indicators of industry of firms with 20K-5M in revenue
 - Industry Effect
 - Time Effect
 - X_{it} = dummies for status
- Standard errors are clustered to correct for correlation within the industries.



- The scatter plot shows that most firms cluster around zero trade shock, suggesting limited exposure for many. Small firms tend to show higher log revenue growth than large firms with in an industry, especially near zero or positive trade shocks. Small firms appear more resilient to trade shocks, especially near zero or positive values. Large firms show lower log revenue growth. They are more likely to be exposed to very large trade shocks. The relationship between trade shock and growth is not strongly linear. Firm size moderates the effect of trade shocks on revenue growth with smaller firms tends to negate the effects of the larger firms.
- Using cleaned panel data with outliers removed the scatter plot shows clear differences across firm sizes. Small firms consistently achieve higher log revenue growth than large firms, especially under low trade shock exposure, with tightly clustered outcomes indicating greater resilience. In contrast, large firms exhibit wider dispersion in both trade shocks and growth, suggesting more heterogeneous and potentially vulnerable responses. Overall, the relationship between trade shocks and revenue growth is not clearly defined.

Estimations

PanelOLS Estimation Summary							PanelOLS Estimation Summary						
Dep. Variable:	log_growth	R-squared:	0.4931	Dep. Variable:	log_growth	R-squared:	0.6620	Dep. Variable:	log_growth	R-squared:	0.6620	Dep. Variable:	log_growth
Estimator:	PanelOLS	R-squared (Between):	0.8801	Estimator:	PanelOLS	R-squared (Between):	0.8350	Estimator:	PanelOLS	R-squared (Between):	0.8350	Estimator:	PanelOLS
No. Observations:	22460	R-squared (Within):	0.4956	No. Observations:	18783	R-squared (Within):	0.6652	No. Observations:	18783	R-squared (Within):	0.6652	No. Observations:	18783
Date:	Fri, Dec 12 2025	R-squared (Overall):	0.7753	Date:	Fri, Dec 12 2025	R-squared (Overall):	0.7690	Date:	Fri, Dec 12 2025	R-squared (Overall):	0.7690	Date:	Fri, Dec 12 2025
Time:	18:24:24	Log-likelihood	-4.418e+04	Time:	18:24:24	Log-likelihood	-2.988e+04	Time:	18:24:24	Log-likelihood	-2.988e+04	Time:	18:24:24
Cov. Estimator:	Clustered	F-statistic:	6782.7	Cov. Estimator:	Clustered	F-statistic:	1.128e+04	Cov. Estimator:	Clustered	F-statistic:	1.128e+04	Cov. Estimator:	Clustered
Entities:	1532	P-value	0.0000	Entities:	1496	P-value	0.0000	Entities:	1496	P-value	0.0000	Entities:	1496
Avg Obs:	14.661	Distribution:	F(3,20915)	Avg Obs:	12.555	Distribution:	F(3,17274)	Avg Obs:	12.555	Distribution:	F(3,17274)	Avg Obs:	12.555
Min Obs:	1.0000	F-statistic (robust):	1894.9	Min Obs:	1.0000	F-statistic (robust):	1376.6	Min Obs:	1.0000	F-statistic (robust):	1376.6	Min Obs:	1.0000
Max Obs:	34.000	P-value	0.0000	Max Obs:	31.000	P-value	0.0000	Max Obs:	31.000	P-value	0.0000	Max Obs:	31.000
Time periods:	11	Distribution:	F(3,20915)	Time periods:	11	Distribution:	F(3,17274)	Time periods:	11	Distribution:	F(3,17274)	Time periods:	11
Avg Obs:	2041.8			Avg Obs:	1707.5			Avg Obs:	1707.5			Avg Obs:	1707.5
Min Obs:	1451.0			Min Obs:	1195.0			Min Obs:	1195.0			Min Obs:	1195.0
Max Obs:	2270.0			Max Obs:	1945.0			Max Obs:	1945.0			Max Obs:	1945.0
Parameter Estimates							Parameter Estimates						
Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI		Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI	
shock	0.0071	0.0017	4.2357	0.0000	0.0038	0.0103	shock	0.0051	0.0014	3.7416	0.0002	0.0024	0.0078
small	4.5418	0.0627	72.400	0.0000	4.4188	4.6647	small	4.4273	0.0722	61.348	0.0000	4.2858	4.5687
trade_small	-0.0089	0.0050	-1.7660	0.0774	-0.0187	0.0010	trade_small	-0.0043	0.0028	-1.5557	0.1198	-0.0098	0.0011
F-test for Poolability: 10.367							F-test for Poolability: 24.874						
P-value: 0.0000							P-value: 0.0000						
Distribution: F(1541,20915)							Distribution: F(1505,17274)						
Included effects: Entity, Time													

- Estimation suggest that higher trade shock is associated with higher log revenue growth, holding other factors constant. The effect size is modest but robust with a p-value of 0%.
- Size of firms matters. Coefficient indicates that small firms, on average, have substantially higher log revenue growth compared to large firms. This aligns with your scatter plots showing small firms clustered at higher growth levels.

- The interaction term is negative coefficient, marginally significant (just above the 5% threshold, but below 10%), suggesting that the interaction between being small and experiencing trade shock slightly reduces growth relative to the baseline. In other words, while small firms generally grow faster, their advantage diminishes somewhat as trade shocks increase.

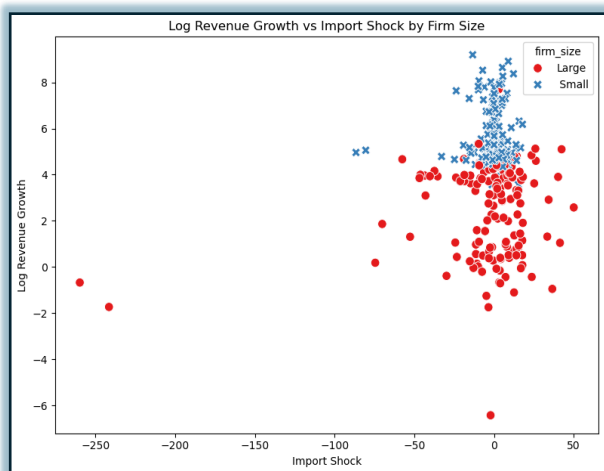


Figure : Long Revenue Growth vs Import Shock by firm size

Small firms (blue crosses) cluster more tightly around the center of the import shock axis and they tend to show higher log revenue growth than large firms. They appear less sensitive to variation in import shock, suggesting resilience or limited exposure.

Large firms (red circles) display a wider spread in both import shock and revenue growth. Some show negative growth under higher import shocks, indicating vulnerability. The dispersion may suggest some large firms may be more exposed to global competition

growth under higher import shocks, indicating vulnerability. The dispersion may suggest some large firms may be more exposed to global competition

Estimations

PanelOLS Estimation Summary						
=====						
Dep. Variable:	log_growth	R-squared:	0.4925			
Estimator:	PanelOLS	R-squared (Between):	0.8778			
No. Observations:	22460	R-squared (Within):	0.4950			
Date:	Fri, Dec 12 2025	R-squared (Overall):	0.7736			
Time:	18:24:25	Log-likelihood	-4.419e+04			
Cov. Estimator:	Clustered	F-statistic:	6766.6			
		P-value	0.0000			
Entities:	1532	Distribution:	F(3,20915)			
Avg Obs:	14.661					
Min Obs:	1.0000					
Max Obs:	34.000	F-statistic (robust):	1669.9			
		P-value	0.0000			
Time periods:	11	Distribution:	F(3,20915)			
Avg Obs:	2041.8					
Min Obs:	1451.0					
Max Obs:	2270.0					
Parameter Estimates						
=====						
Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI	
shock2	0.0008	0.0018	0.4755	0.6344	-0.0026	0.0043
small	4.5124	0.0647	69.795	0.0000	4.3856	4.6391
trade_small	-0.0023	0.0047	-0.4936	0.6216	-0.0115	0.0069
=====						
F-test for Poolability: 10.338						
P-value: 0.0000						
Distribution: F(1541,20915)						
Included effects: Entity, Time						

PanelOLS Estimation Summary						
=====						
Dep. Variable:	log_growth	R-squared:	0.6620			
Estimator:	PanelOLS	R-squared (Between):	0.8340			
No. Observations:	18783	R-squared (Within):	0.6653			
Date:	Fri, Dec 12 2025	R-squared (Overall):	0.7685			
Time:	18:24:25	Log-likelihood	-2.988e+04			
Cov. Estimator:	Clustered	F-statistic:	1.128e+04			
		P-value	0.0000			
Entities:	1496	Distribution:	F(3,17274)			
Avg Obs:	12.555					
Min Obs:	1.0000					
Max Obs:	31.000	F-statistic (robust):	1372.9			
		P-value	0.0000			
Time periods:	11	Distribution:	F(3,17274)			
Avg Obs:	1707.5					
Min Obs:	1195.0					
Max Obs:	1945.0					
Parameter Estimates						
=====						
Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI	
shock2	0.0045	0.0011	4.1194	0.0000	0.0024	0.0067
small	4.4115	0.0734	60.073	0.0000	4.2676	4.5554
trade_small	-0.0012	0.0024	-0.5090	0.6108	-0.0060	0.0036
=====						
F-test for Poolability: 24.877						
P-value: 0.0000						
Distribution: F(1505,17274)						
Included effects: Entity, Time						

Same results as with the total trade shocks. Conclusion: Small firms enjoy a strong growth premium overall, yet their sensitivity to trade shocks may be slightly negative compared to large firms.

6. Team Collaboration in Analysis

- All members actively participated in each task, supporting one another when blockers arose.
- Challenges included large file sizes, schema inconsistencies, and missing data.
- Issues were resolved quickly through group communication, often within hours of being flagged.
- This collaborative approach ensured that each research question was fully analyzed and contributed to the unified report.

Conclusions

This study examined how international trade shocks and exchange rate fluctuations influenced Canadian industries between 2013 and 2023. By merging firm-level financial data with industry trade exposure and exchange rate observations, we found that revenue growth is modestly linked to exports and currency movements, while expenditure growth remains largely independent. Trade flows show structural independence, with exports and imports driven by different sectors, and firm size emerges as a critical factor in resilience- smaller firms are consistently more vulnerable to shocks than larger ones.

At the same time, the analysis faced important data limitations. Revenue data was not consistently available across provinces, and expense data lacked corporate-level detail, making it difficult to merge financial datasets due to differences in granularity. Trade exposure data was incomplete, with only a subset of industries covered. These gaps likely reflect both the sensitivity of financial information and restrictions on open-source availability. As a result, the conclusions are based only on the datasets accessible online.

In addition, the analysis faced structural data challenges. While revenue data provided province-level detail, the incorporation status variable was incomplete - coded only as levels 1 and 2, whereas documentation suggests three levels should exist. Expense files, meanwhile, only reported two levels of incorporation, creating inconsistencies across datasets. These mismatches in granularity and classification made merging financial data difficult and limited the scope of comparisons. We proceeded with the available data, but emphasize that the conclusions are based on partial open-source datasets and are intended for learning purposes only, not for definitive policy or financial decision-making

Final Submission Checklist

1. Report (PDF or DOCX, ≤ 15 pages).
2. Jupyter Notebook (.ipynb).
3. Supporting files (CSV/XLS/TXT if needed).
4. References.
5. Submit all files together in the Group Assignment Dropbox.

Reference of Data source

Financial Performance Data –

- 2012 - <https://open.canada.ca/data/en/dataset/13544e75-e25b-4883-bca7-418b37a4331d>
- 2013 - <https://open.canada.ca/data/en/dataset/de4dc179-97d0-4323-98d0-0b246d01e7f1>
- 2014 - <https://open.canada.ca/data/en/dataset/7c89bb23-ba7c-4617-b411-7e82f5f853a1>
- 2015 - <https://open.canada.ca/data/en/dataset/f559526c-40f9-4c5d-bced-0f7a41f278b9>

2016 - <https://open.canada.ca/data/en/dataset/37653ec6-9c7d-4e74-82b0-299d22530519>

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2023 - <https://open.canada.ca/data/en/dataset/16b7eeb0-9a97-46c0-af3d-7f2114dedf5e>

Trade exposure by industry-

<https://open.canada.ca/data/en/dataset/79438be5-02be-43ec-aa3f-65cdb47b084d/resource/f248af56-bc40-4e37-8d97-257d91b4bebb>

Total Trade/ GDP

<https://www150.statcan.gc.ca/t1/tb11/en/tv.action?pid=1210017101&pickMembers%5B0%5D=2.1&cubeTimeFrame.startYear=2012&cubeTimeFrame.endYear=2024&referencePeriods=20120101%2C20240101>

Exchange rate

<https://www150.statcan.gc.ca/t1/tb11/en/tv.action?pid=3310003601>

Image ref.

https://www.shutterstock.com/image-vector/canada-economy-flag-rising-financial-graph-2658654415?dd_referrer=https%3A%2F%2Fwww.google.com%2F